

Ahmeduddin Mohammed

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Professional Summary

ML Engineer building production ML pipelines, LLM agent systems, and causal inference engines. Shipped systems with AUC 0.9868, 27ms p95 latency, 92% test coverage, and live GCP deployments.

Technical Skills

ML & Deep Learning: PyTorch, LSTM, Transformer, XGBoost, scikit-learn, SHAP, Optuna
Causal & LLM: DoWhy, EconML, CausalForestDML, LangGraph, LangChain, GPT-4o-mini, RAG
Data & Vector: Pandas, NumPy, ChromaDB, sentence-transformers, dbt, PostgreSQL, Redis
MLOps & Cloud: MLflow, Docker, GCP Cloud Run, GitHub Actions, HuggingFace Spaces, pytest
Backend & APIs: FastAPI, SQLAlchemy, Pydantic v2, JWT, Alembic, REST APIs
Languages: Python 3.11, TypeScript, SQL, JavaScript, Bash, Java

Professional Experience

Jr. ML Engineer

Jul 2025 – Present

FL Tech Solutions LLC

Florida, United States (Remote)

- Validated 50K–200K-row datasets with Pandas and SQL; MLflow 3+ A/B runs improved accuracy by **~15%**.
- Evaluated LangChain, ChromaDB, sentence-transformers; cut decision time **~20%**, pipeline errors **~30%**.
- Maintained model cards across 5+ systems; 10+ reviews cut onboarding and resolution time by **~25%**.

Python & Data Science Instructor

Aug 2023 – Dec 2023

Full Stack Academy

Hyderabad, India (Remote)

- Taught **20+ students** Python, OOP, and algorithms using Pandas, NumPy, and Matplotlib over 5 months.
- Designed **15+ exercises** on EDA and data wrangling using Pandas, NumPy, Seaborn, and Matplotlib.
- Mentored **3+ mini-projects** on preprocessing and feature engineering for data science tasks.

Data Science Intern

Jun 2022 – Jul 2022

Delta Sigma Technologies

Hyderabad, India (Remote)

- Built pipelines for **100K+ records** using Pandas and NumPy with imputation and encoding.
- Ran **50+ tuning experiments** via grid search and CV, lifting model accuracy from **76% to 89%**.
- EDA with Seaborn and Matplotlib on distributions, correlations, and class imbalances for review.

Projects

SessionScout – Real-Time E-Commerce Conversion Scoring System

PyTorch, LSTM, Transformer, XGBoost, FastAPI, Redis, Docker, HuggingFace Spaces

- 4-model cascade (LR → XGBoost → LSTM → Transformer); **AUC 0.9868, 27ms p95**, Redis caching.
- 121 features** (session, sequence, device); gating routed **60%** to cheaper models, cutting cost.
- Deployed on HuggingFace Spaces with GitHub Actions CI/CD; **121 unit tests, 93% coverage**.

LiftLab – Causal Inference Engine for Promotion Uplift

DoWhy, EconML, CausalForestDML, SHAP, MLflow, PSI, MMD, Streamlit

- 5 estimators (PSM, DiD, T/X-Learner, CausalForest) on **14M rows**; segments showed **3.5x CATE lift**.
- CausalForestDML honest splitting gave confidence intervals; MMD at **0.136** flagged drift.
- SHAP validated cross-estimator consistency; **53+ tests, 81% coverage** with MLflow tracking.

LoanLens – AI Credit Risk Explainer with CFPB Reg-B Compliance

XGBoost, SHAP, LangChain, ChromaDB, GPT-4o-mini, dbt, PostgreSQL, MLflow

- Scored **307K applications**; XGBoost **AUC 0.77, KS 0.41**; SHAP mapped to 90+ CFPB Reg-B codes via dbt.
- 9,977 chunks** in ChromaDB; GPT-4o-mini generates notices in **under 4s**; grounding score **1.0**.
- PSI drift monitoring with Slack alerting; **96 tests, 89% coverage**.

CareAgent – Multi-Agent Healthcare Provider Quality Scoring

LangGraph, LangChain, GPT-4o-mini, Isolation Forest, GCP Cloud Run, PostgreSQL

- Orchestrated **5 agents** via LangGraph StateGraph; automated scoring of **10,000 CMS Medicare providers**.
- 0.36s** pipeline; Isolation Forest flagged **3% anomaly rate**; routing skips low-quality stages.
- Deployed on GCP Cloud Run with LangSmith tracing; **99 tests, 92% coverage**.

Education

New Jersey Institute of Technology

Jan 2024 – May 2025

Master of Science in Computer Science

Newark, NJ

Muffakhham Jah College of Engineering & Technology, Osmania University

Aug 2019 – Jul 2023

Bachelor of Engineering in Information Technology

Hyderabad, India